

Jaume **Ivars Grimalt**

AI Engineer · Computer Vision

Valencia, Spain · open to relocation

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Profile

I build vision models end to end: I design them, train them from scratch at scale, measure them against human experts and ship them as production services. At Mycrospace that took high-density colony detection from my Master's thesis to a product **tested in 5 labs**. I care about what decides whether a model holds up outside the benchmark: label quality, error analysis and inference cost.

Technical Skills

Computer Vision	Object detection, segmentation, density estimation, classification; ConvNeXt, YOLO and DETR-style models
Training & Eval	PyTorch (DDP, AMP, torch.compile), timm, Hugging Face, multi-GPU on HPC (SLURM); cross-validation, ensembles
Data	Weakly-supervised labelling tools, annotation protocols with domain experts, label versioning, error analysis
Deployment	ONNX, Docker, GCP (Cloud Run, Cloud SQL, GCS), BullMQ/Redis queues, CI/CD, inference profiling
Software	Python, TypeScript, C++ (working knowledge); NestJS, PostgreSQL, Next.js, React Native/Expo

Experience

Mycrospace · Co-Founder & CTO — *Valencia, Spain* · *Jan 2024 – Sep 2026*

- Computer vision for microbiology. Started in 2024 as my Master's thesis; incorporated in January 2026.
- **Detector designed and trained from scratch:** ConvNeXt backbone with ideas from YOLO and DQ-DETR, handling 10 to 2,000 objects per image. **62.4 mAP50, 42.5 mAP50-95**; validated against expert microbiologists at **90% precision and 75% recall**.
- **Training at scale:** **12,500+ GPU-hours** on the Leonardo supercomputer under two EuroHPC AI Factory allocations. Own framework: 4x A100 with DDP, AMP and torch.compile under SLURM, chained jobs resumed from checkpoint, cross-validation and ensembles with weighted box fusion.
- **Data and labels:** **1,500 of 5,000 images** annotated under protocols written with microbiologists. Built weakly-supervised labelling tools (click → crop → 4-channel CNN with a Gaussian heatmap → bbox regression) and annotation versioning with k-fold image scoring, so the model improves as labels do.
- **Production:** containerised inference with BullMQ/Redis queue workers on GCP (Cloud Run, Cloud SQL, GCS), signed-URL uploads, real-time results over SSE/WebSocket.
- **Product:** NestJS/PostgreSQL backend, Next.js web client, React Native/Expo bench app and Stripe billing.

MIALAB, Medical Imaging Analysis Laboratory (UPV) · Research Engineer, part-time — *Valencia, Spain* · *May 2025 – Jan 2026*

- Volumetric brain MRI segmentation with nnU-Net, comparing multi-scale feature-extraction strategies.
- VolBrain has processed **700k+ MRI volumes**, so inference cost scales: profiled the pipeline from NIfTI ingest to voxel-level prediction to find where the time went.

Earlier roles

Tapstar, Full-Stack Engineer (freelance)	QR cards product: generation, back office, scan analytics	*Jan 2025 – May 2025*
Vesta-Z, Data Engineer	Cloud product features, cross-provider matching schema, LLM bot	*Jul 2023 – Feb 2025*
Neurocatching, Data Scientist Intern	Eye-gaze model training for neuromarketing studies	*Feb 2023 – Aug 2023*
Sciling, Backend Developer Intern	Conversational systems on GPT-3-era LLMs, applied NLP	*Jun 2022 – Sep 2022*

Project

ConvNeXt-V1 reproduced from scratch · github.com/jaume2000/convnext_v1 — *Leonardo Booster* · *2026*

- ConvNeXt-Tiny (28M parameters) trained on ImageNet-1k for 300 epochs on 4x A100 with my own framework: **81.0% top-1** vs 82.1% in the paper. A baseline I own, now used to study parameter redundancy and residual blocks as Euler-solver steps.

Publications

- Auto-N2N: Self-supervised MRI Denoising using Synthetic Pair Generation and Mixture of Experts. S. Morell-Ortega, A. Mengliev, A. Gonzalez-Cebrian, **J. Ivars Grimalt**, P. Coupé, J. V. Manjón. Preprint, 2026.
- Detecting Cytopathic Effect in Cell Culture with Computer Vision. W. Ragazzo, **J. Ivars Grimalt**. Under review.
- One Block, 10,000 Steps: ConvNeXt Stages as Nearly Autonomous Vector Fields. **J. Ivars Grimalt**. In preparation.

Education & Awards

M.Sc. in Artificial Intelligence, Pattern Recognition & Digital Imaging, UPV. GPA 8.7/10; thesis awarded Honours	*2023 – 2024*
B.Sc. in Computer Engineering, UPV. GPA 8.3/10; honour mentions in Machine Learning and Statistics	*2019 – 2023*
Technology Management, CDTM (TUM/LMU Munich), Valencia campus. Add-on programme, 16 students selected	*2024*
1st Prize, IdeasUPV Startup Competition with Mycrospace, 2,500 EUR grant	*2024*

Languages Spanish (native), Catalan (native), English (C1)